

SHIV TANEJA

LinkedIn — GitHub

Email : shivtaneja16@gmail.com

Mobile : +91-6397054423

PROFESSIONAL SUMMARY

Innovative AI Developer with hands-on experience designing and building enterprise-grade AI solutions. Specialized in developing robust systems using **Python, LLMs, Retrieval-Augmented Generation (RAG), and Agentic Workflows**. Proven track record in optimizing data extraction pipelines for diverse document types, implementing OCR solutions, and iterating on indexing/retrieval mechanisms to enhance system performance. Strong focus on **prompt engineering, agentic design patterns**, and delivering scalable, production-ready GenAI architectures.

EXPERIENCE

- **Tata Consultancy Services (TCS)** New Delhi, India
Nov 2023 - Present
Developer
 - **Agentic AI & LLM Frameworks:** Designed and deployed Agentic AI applications using Python, LangChain, and LangGraph to orchestrate multi-step reasoning workflows and enterprise automation tools. Developed ReAct-based agents integrating retrieval tools, reasoning chains, and dynamic task decomposition.
 - **Modular RAG & Retrieval Systems:** Architected production-grade Modular RAG pipelines incorporating hybrid search, semantic retrieval, chunking strategies, and re-ranking mechanisms. Designed indexing strategies using FAISS and vector databases to optimize retrieval latency (TTFT < 200ms). Evaluated retrieval quality using RAGAS evaluation metrics to reduce hallucination and improve answer grounding.
 - **Fine-Tuning & Model Optimization:** Fine-tuned SLMs/LLMs using PEFT methods including LoRA and QLoRA for domain-specific enterprise use cases. Optimized prompt strategies and inference pipelines, achieving 34% automation improvement across internal workflows.
 - **Large-Scale Data Handling & OCR:** Designed document ingestion pipelines integrating OCR systems with Docling for high-accuracy text extraction from complex, unstructured formats. Reduced manual document processing workload by 40% through automation and AI-driven extraction systems.
 - **LLMOps, Deployment & Governance:** Implemented LLMOps practices including experiment tracking (MLflow), containerization (Docker), and scalable cloud deployment. Incorporated responsible AI safeguards including prompt injection fallback strategies and mitigation of unsafe model behaviors.

PROJECTS

- **Industry Policy chatbot:** An intelligent chatbot that answers questions about Karnataka's New Industrial Policy 2025–2030 using a **Hybrid RAG (Retrieval-Augmented Generation) pipeline**. It combines **sparse keyword search, dense semantic search, and cross-encoder reranking** to surface the most relevant policy excerpts, then uses a **ReAct Agent** to generate accurate, context-grounded answers.

EDUCATION

- **VIT Bhopal** Bhopal, India
2019-2023
B.Tech in Computer Science and Engineering (Minor: AI & ML); GPA: 8.53

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Generative AI Leader Certification
- AWS Certified Cloud Practitioner

SKILLS

- **ML & Deep Learning Frameworks:** TensorFlow, PyTorch, Scikit-learn
- **Generative AI & LLM Stack:** Python, LangChain, LangGraph, RAGAS, LoRA, QLoRA, PEFT, FAISS, Pinecone, Weaviate, async programming
- **MLOps & Deployment Tools:** MLflow, Docker, AWS, Streamlit, Git